

AN ADDITIVE GIRAUD THEOREM

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Introduction

For V a cocomplete locally finitely presentable closed category [6] many of the familiar processes of topos theory may be V -enriched. For example, the concept of Grothendieck topology, the associated sheaf construction (as a two-stage process) and change of universe are available. These features of V -topos theory are described in Section 1 (cf. [1] and [7]).

In Section 2 we consider canonical additive topologies and some consequences. Necessary and sufficient conditions are then given for a V -category E to be equivalent to a V -topos for certain additive closed categories V . More precisely, this Giraud-style characterisation of V -topoi is proved for the convolutions (see [2]) $V = [M, Ab]$ where Ab denotes the closed category of abelian groups and M is a small symmetric compact closed Ab -category (see [3] and [5]).

1. The associated-sheaf construction

Let $V = (V_0, \otimes, I, [-, -], \dots)$ be a cocomplete locally finitely presentable symmetric monoidal closed category with small strong set of generators H , as defined in Kelly [6] § 5. Henceforth all categories, functors and so on, will be assumed to be relative to this V unless otherwise indicated.

(1.1) Definition: A Grothendieck $\mathcal{H}$ -topology on a category C is defined by giving, for each object C of C , a set $J(C)$ of subfunctors of the representable functor $C(-, C)$, called the covering cribles of the topology, satisfying the axioms:

G1. For each $C \in C$, $1: C(-, C) \rightarrow C(-, C)$ is in $J(C)$,

f NOT SCRIPT G2. For all $x \in \mathcal{H}$, $R \in J(C)$ and $f: C(-, D) \rightarrow [x, C(-, C)]$, $f^{-1}[x, R]$ is in $J(D)$,

G3. If $R \in J(C)$ and S is a crible on C such that, for each $f: D \rightarrow C$ in R , we have $f^{-1}S \in J(D)$, then S is in $J(C)$. $\square$

A category equipped with such a topology is called an $\mathcal{H}$ -site.

An immediate consequence of these axioms is the following:

(1.2) Lemma. The intersection of two covering cribles is a covering crible and any crible through which a covering crible factors is a covering crible. $\square$

Given an $\mathcal{H}$ -site (C, J) call a functor P in $P = [C^{op}, V]$ a sheaf (respectively separated presheaf) if the induced map

$$P(C(-, C), P) \rightarrow P(R, P)$$

is an isomorphism (respectively monomorphism) for all R in $J(C)$ and C in C . A left-exact left retract of a functor category $[C^{op}, V]$ with C small is called a V -topos.

We now show that each small $\mathcal{H}$ -site (C, J) gives rise to a V -topos, namely the category of sheaves on (C, J) .

(1.3) Construction. For a small $\mathcal{H}$ -site (C, J) , there is a (V -enriched) left-exact endofunctor L defined on $P = [C^{op}, V]$ as follows. For P in P and C in C , put

$$LP(C) = \operatorname{colim}_{R \in J(C)} P(R, P).$$

The conditions for a small $\mathcal{H}$ -site C give rise to factorisations:

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$$\begin{array}{ccc}
 \sum_{x \in X} V_0(x, C(D, C)) \cdot x & & \\
 \downarrow & \searrow & \\
 C(D, C) & \dashrightarrow & \lim_{R} \operatorname{colim}_{S} P(S, R) \\
 \parallel \downarrow & & \downarrow \\
 P(C(-, D), C(-, C)) & \longrightarrow & \lim_{R} \operatorname{colim}_{S} P(S, C(-, C)) \\
 & \searrow & \parallel \downarrow \\
 & & \operatorname{colim}_{S} P(S, C(-, C)),
 \end{array}$$

where the exterior commutes by axiom G2 and the isomorphisms follow from the V -Yoneda lemma and the fact that $J(C)$ is connected since it is cofiltered by (1.2). Next, by the V -Yoneda lemma applied to P in P , the above factorisations give maps:

$$\begin{array}{ccc}
 C(D, C) & \xrightarrow{\quad \quad \quad} & \int_P [LP(C), LP(D)] \\
 & \searrow & \swarrow \cong \\
 & \lim_{R} \lim_{S} P(S, R) &
 \end{array}$$

which are easily computed to be the components of a V -functor L' from C^{op} to P' where P' is the V -category:

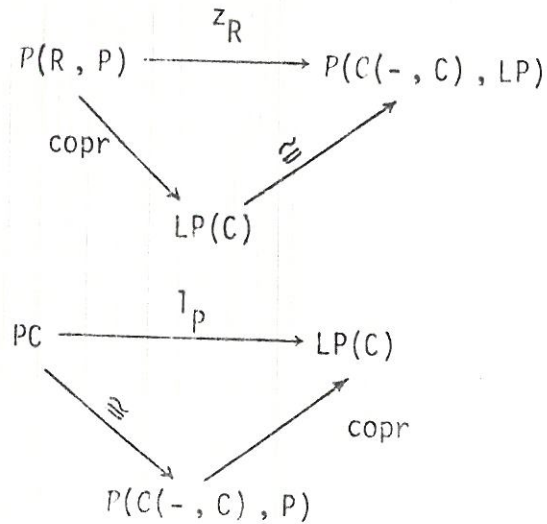
$$\{Q \in [P, V]; \int_P [QP, HP] \text{ exists for all } H \text{ in } [P, V]\}.$$

But the evaluation functor Ev from $P' \otimes P$ to V is a V -functor so we have the composite

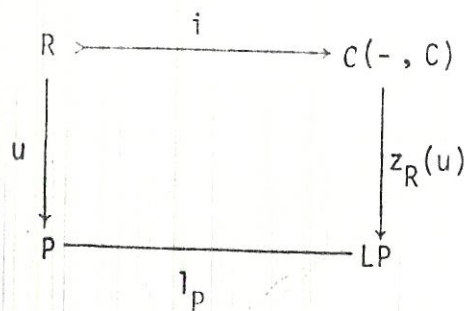
$$C^{op} \otimes P \xrightarrow{L' \otimes 1} P' \otimes P \xrightarrow{Ev} V;$$

that is, we obtain a V -endofunctor L on P which is clearly left-exact in the sense that it preserves finite conical limits and cotensoring by the finitely presentable objects of V (since L is defined by a filtered colimit). $\square$

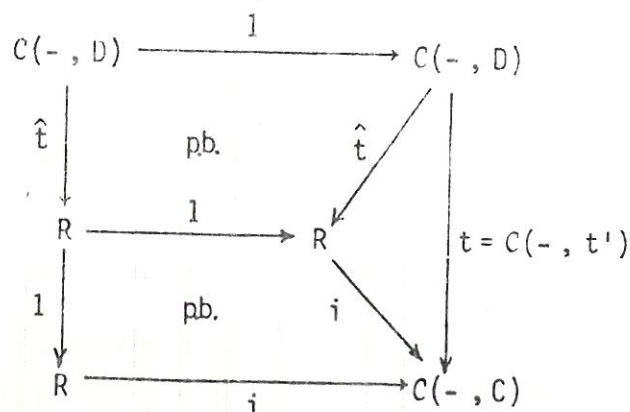
Next we define transformations z_R and l_P by means of the V -Yoneda lemma and the diagrams:



(1.4) Lemma. Given R in $J(C)$ and u in $P_0(R, P)$ then $z_R(u)$ is the unique morphism making the following diagram commute:



Proof. Uniqueness of $z_R(u)$ is immediate from the V -Yoneda lemma, the assumption that I is finitely presentable and the definition of $LP(C)$ as a filtered colimit. In order to show $z_R(u) \cdot i = l_P \cdot u$ note that the representables $C(-, D)$ generate so it suffices to show $z_R(u) \cdot t = l_P \cdot u \cdot \hat{t}$ for all "test" maps $\hat{t}: C(-, D) \rightarrow R$ where $t = i_R \cdot \hat{t}$. But the exterior of the diagram



is a pullback since i is a monomorphism. Thus, by the definition of LP on morphisms, the exterior of the following diagram commutes:

$$\begin{array}{ccccc}
 P(R, P) & \xrightarrow{z_R} & P(C(-, C), LP) & \xrightarrow{\cong} & LP(C) \\
 \downarrow P(\hat{t}, P) & & \downarrow P(t, LP) & & \downarrow LPt' \\
 P(C(-, D), P) & \xrightarrow{P(1, 1_P)} & P(C(-, D), LP) & \xrightarrow{\cong} & LP(D)
 \end{array}$$

Thus the result follows from considering $u \in P_0(R, P)$ in this diagram. $\square$

(1.5) Lemma. For all $f: C(-, C) \rightarrow LP$ there exists R in $J(C)$ and $u: R \rightarrow P$ with $z_R(u) = f$.

Proof. Apply $V_0(I, -)$ to the definition of z_R and use the facts that I is finitely presentable and $LP(C)$ is a filtered colimit. $\square$

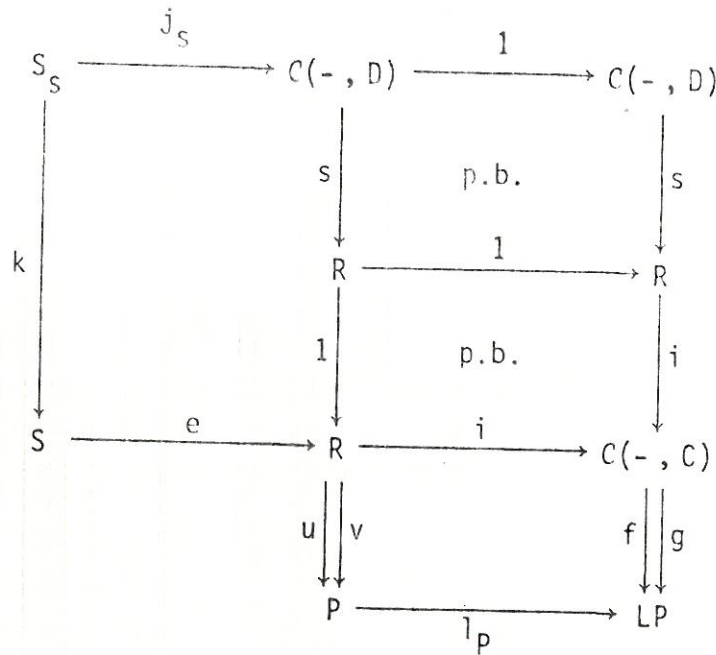
(1.6) Lemma. For all presheaves P , LP is a separated presheaf.

Proof. Since L preserves cotensoring by the objects of $\mathcal{A}$ it suffices to prove $P_0(i_R, P)$ is an injection for all covering cribles i_R into $C(-, C)$. Let i_R equalise f and g from $C(-, C)$ to LP . Using Lemma (1.5) and intersecting the two resultant cribles, we obtain that

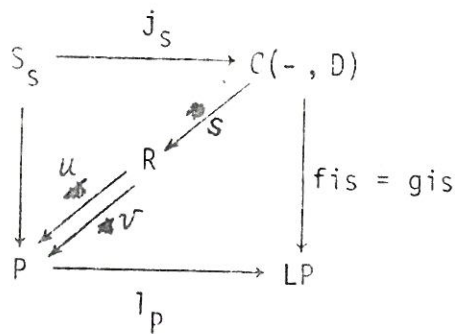
$$\begin{array}{ccc}
 R' & \xrightarrow{i} & C(-, C) \\
 \downarrow u \quad \downarrow v & & \downarrow f \quad \downarrow g \\
 P & \xrightarrow{1_P} & LP
 \end{array}$$

8.

commute, where $fi_{R'} = gi_{R'}$ since $fi_R = gi_R$. Now write R for R' and let $e: S \rightarrow R$ be the equaliser of u and v . To prove $f = g$ it suffices, by Lemma (1.4), to prove i covers. Consider the following diagram:



We have that i covers so, by axiom G3 and Lemma (1.2), it suffices to produce j_s covering for an arbitrary s from $C(-, D)$ to R . But, $fi = gi$ so $fis = gis$ so $usj_s = vsj_s$ for some covering crible j_s from S_s to $C(-, D)$ by Lemma (1.5):



please keep labels vertical

Thus the map k exists by the equaliser property of e from S to R . $\square$

(1.7) Corollary. A presheaf P is separated if and only if l_P is a monomorphism.

Proof. If P is separated then we have the following filtered colimit of monomorphisms:

$$\begin{array}{ccc}
 P(C(-, C), P) & \xrightarrow{P(i, P)} & P(R, P) \\
 \downarrow \text{at} & & \downarrow \\
 PC & \xrightarrow{l_P} & LP(C)
 \end{array}$$

If $l_P: P \rightarrow LP$ is a monomorphic transformation then P is separated since it is a subobject of a separated presheaf. $\square$

(1.8) Lemma. If P is a separated presheaf and v maps $C(-, C)$ to LP then the pullback of l_P along v is a covering crible.

Proof. By Corollary (1.7) the pullback of l_P along v is a monomorphism. The result now follows on combining Lemmas (1.4) and (1.5) and then using Lemma (1.2) together with the universal property of pullbacks. $\square$

(1.9) Lemma. If P is a separated presheaf then LP is a sheaf.

Proof. If P is separated then so is $[x, P]$ for all x in $\mathcal{H}$ by Corollary (1.7) and the fact that cotensoring by the objects of $\mathcal{H}$ is preserved by L . Thus we require that $P_0(i_R, [x, LP])$ be a surjection for all R in $J(C)$. Replace $[x, P]$ by P and suppose w in

$P_0(R, LP)$. Consider the diagram

$$\begin{array}{ccccc}
 S & \xrightarrow{h} & R & \xrightarrow{i} & C(-, C) \\
 \downarrow u & & \downarrow w & \nearrow z_S(u) & \\
 P & \xrightarrow{l_P} & LP & &
 \end{array}
 \quad \text{p.b.}$$

It follows from axiom G3 and Lemma (1.8) that ih covers. Thus, by Lemma (1.4), we obtain $z_S(u)$ such that diagram $*$ commutes by Lemmas (1.8) and (1.4), the fact that LP is separated and the generating property of the representables. $\square$

(1.10) Corollary. A presheaf P is a sheaf if and only if l_P is an isomorphism.

Proof. If P is a sheaf then l_P is a colimit of isomorphisms. Conversely, if l_P is an isomorphism then P is a sheaf since $P \cong LP$ and LP is separated (so that P is separated with $P \cong LP$). $\square$

These results are summarised as follows

(1.11) Theorem. Given a small Grothendieck $\mathcal{M}$ -site (C, J) there is a V -endofunctor L on $P = [C^{op}, V]$ whose value on objects is given by

$$LP(C) = \operatorname{colim}_{R \in J(C)} P(R, P)$$

and L is left exact in that it preserves finite conical limits and cotensoring by the finitely presentable objects of V . Moreover L^2 is the reflection of P into the category $Sh(C, V)$ of all V -valued sheaves on (C, J) .

Proof. It remains to prove L^2 is left adjoint to the inclusion of $Sh(C, V)$ into $[C^{op}, V]$. Since we already know that L^2 is a V -functor with values in $Sh(C, V)$ it suffices to prove each transformation f from a presheaf P to a sheaf F lifts uniquely along l_P (for then $l_{LP} \cdot l_P$ is the unit of the reflection). Firstly

$$\begin{array}{ccc}
 P & \xrightarrow{l_P} & LP \\
 f \downarrow & & \downarrow Lf \\
 F & \xrightarrow[l_F]{\cong} & LF
 \end{array}$$

commutes, where l_F is an isomorphism by Corollary (1.10).

Secondly let g, h be two distinct liftings of f along l_P :

$$\begin{array}{ccc}
 P & \xrightarrow{l_P} & LP \\
 f \searrow & & \nearrow g, h \\
 & F &
 \end{array}$$

*please keep
labels vertical*

Let $t: C(-, C) \rightarrow LP$ be such that $gt \neq ht$; then Lemmas (1.4) and (1.5) can clearly be combined to deduce a contradiction. $\square$

Finally we mention that there is an obvious change-of-universe procedure for $\mathcal{V} = \text{Lex}(\mathcal{V}_f, \text{Set})$. This procedure is given by replacing Set by a larger universe SET with respect to which $\mathcal{V}$ and Set are both small, then considering $\hat{\mathcal{V}} = \text{Lex}(\mathcal{V}_f, \text{SET})$. Because limits and filtered colimits are formed pointwise in $\hat{\mathcal{V}}$ we deduce the following result.

(1.12) Corollary. Given a $\hat{\mathcal{V}}$ -small H -site $(\mathcal{C}, J)$ there exists a left $\hat{\mathcal{V}}$ -adjoint to the inclusion of $\text{Sh}(\mathcal{C}, \mathcal{V})$ into $[\mathcal{C}^{\text{op}}, \mathcal{V}]$ if $J(\mathcal{C})$ contains a $\mathcal{V}$ -small initial directed (cofiltered) subcategory for each $\mathcal{C}$ in $\mathcal{C}$. $\square$

2. The additive Giraud theorem

Henceforth we shall suppose that the base category $\mathcal{V}$ is also additive and that $\mathcal{E}$ is a $\hat{\mathcal{V}}$ -small $\mathcal{V}$ -category.

(2.1) Definition. $\mathcal{E}$ is called completely regular if it satisfies the following three conditions:

- (i) $\mathcal{E}$ is finitely complete and has all small directed colimits (that is, colimits indexed by directed posets);
- (ii) strong epimorphisms and small directed colimits are universal in $\mathcal{E}$;
- (iii) cotensoring with each object in $\mathcal{H}$ preserves strong epimorphisms and small directed colimits. $\square$

(2.2) Proposition. If $\mathcal{E}$ is completely regular then there is induced on $\mathcal{E}$ a Grothendieck $\mathcal{H}$ -topology for which the covering cibles are those subfunctors R of $\mathcal{E}(-, E)$ through which some morphism of the form

$$\operatorname{colim}_{\phi \in \Phi} \mathcal{E}(-, E_{\phi}) \longrightarrow \mathcal{E}(-, E)$$

factors, where Φ is a small directed poset and

$$\operatorname{colim}_{\phi \in \Phi} E_{\phi} \longrightarrow E$$

is a strong epimorphism in $\mathcal{E}$.

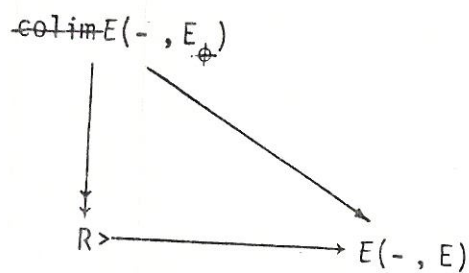
Proof. Axioms G1 and G2 are clearly satisfied, while axiom G3 follows from the additivity of $\mathcal{E}$. $\square$

In a completely regular category E each strong epimorphism is a coequaliser. For a proof of this fact (due to A. Joyal) see [8] Corollary 4.

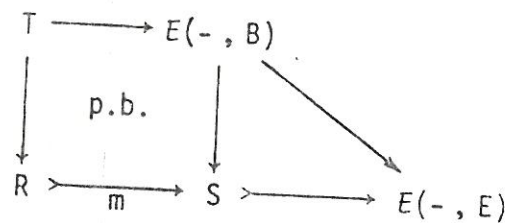
(2.3) Proposition. The representable functors $F = E(-, D) : E^{op} \rightarrow U$ are sheaves for the above topology.

Proof. Clearly the "basic" covering cibles given by the strong-epimorphism, monomorphism factorisations

$E(-, E_\lambda)$



are closed under pullback and have the representables orthogonal to them by regularity of E . Let $S \twoheadrightarrow E(-, E)$ contain such a crible R and let $f : S \rightarrow F$ be a morphism with $F = E(-, D)$. Consider the following diagram



where B is an arbitrary object of E . The exterior is a pullback since $S \twoheadrightarrow E(-, E)$ is a monomorphism; thus T always covers, so

$$\begin{array}{ccc}
 S & \xrightarrow{\quad} & E(-, E) \\
 f \downarrow & & \uparrow g \\
 & & E(-, D)
 \end{array}$$

commutes, where g is the unique extension of f . $\square$

(2.4) Comparison lemma. Let E be completely regular and let $I: C \hookrightarrow E$ be the inclusion of a small strongly generating set which is closed under finite limits and subobjects in E . Give C the restriction topology from E in which R is a basic covering crible if it comes from a strong-epimorphism factorisation of the form

$C(-, C_\lambda)$
 $\downarrow$
 $C(-, C)$

$$\begin{array}{ccc}
 \text{colim } C(-, C_\lambda) & & \\
 \downarrow & \searrow & \\
 R & \xrightarrow{\quad} & C(-, C)
 \end{array}$$

where Λ is a small set and directed set and

$$C_\lambda \rightarrow IC$$

$$\text{colim } IC_\lambda \rightarrow IC$$

is a strong epimorphism in E . Then $Sh(E, V)$ is V -equivalent to $Sh(C, V)$.

Proof. Denote by J_b the basic covering cibles of C and E . Clearly we have an initial functor from $J_b(C)$ to $J_b(IC)$ for each object C in C . Also the diagram

$$\begin{array}{ccc}
 J_b(C) & \longrightarrow & [C^{op}, V] \\
 \downarrow & & \downarrow \\
 J_b(IC) & \longrightarrow & [E^{op}, V]
 \end{array}$$

commutes to within isomorphism by the left-exactness of $\exists_I$ (which follows from [6] Theorem (6.12)). This gives both that $I^*L \cong LI^*$ and that I^* applied to a sheaf is again a sheaf (thus inducing I^S in the diagram below). But we also have that $J(E)$ has a small initial directed subset by the hypotheses on C , thus $L^2: [E^{op}, V] \rightarrow Sh(E, V)$ exists by (1.12), and we obtain the induced adjunction $I_S \dashv I^S$ in the diagram:

$$\begin{array}{ccc}
 [E^{op}, V] & \begin{array}{c} \xleftarrow{\exists_I} \\ \xrightarrow{I^*} \end{array} & [C^{op}, V] \\
 \begin{array}{c} \downarrow L^2 \\ \uparrow L^2 \end{array} & & \begin{array}{c} \downarrow L^2 \\ \uparrow L^2 \end{array} \\
 Sh(E, V) & \begin{array}{c} \xleftarrow{I_S} \\ \xrightarrow{I^S} \end{array} & Sh(C, V)
 \end{array}$$

From $I^*L \cong LI^*$ we now deduce $I^S I_S \cong 1$ so that it remains to show I^S reflects isomorphisms. But the counit of $I_S \dashv I^S$ is simply the reflection into $Sh(E, V)$ of the canonical morphism:

$$\int_C E(-, IC) \otimes FIC \longrightarrow F.$$

Thus the result follows from the V -Yoneda lemma

$$F \cong \int_E E(-, E) \otimes FE$$

and the fact that the morphism

$$\int^C E(-, IC) \otimes E(IC, E) \longrightarrow E(-, E)$$

is a strong epimorphism after L^2 is applied, (since $I: C \subset E$ is strongly generating, the set of all maps $IC \longrightarrow E$, C in C , is covering). $\square$

(2.5) Definition. Call a pair $X \rightrightarrows Y$ of morphisms in $\mathcal{V}$ a congruence in $\mathcal{V}$ if there exists a morphism $Y \longrightarrow Z$ in $\mathcal{V}$ whose pullback with itself is the original pair. Call a pair $D \rightrightarrows E$ of morphisms in E a congruence in E if the induced pair $E(B, D) \rightrightarrows E(B, E)$ is a congruence in $\mathcal{V}$ for all B in E . $\square$

(2.6) Definition. Call E completely exact if it is completely regular and each congruence in E is the kernel pair of some morphism in E . $\square$

It follows that if E is completely exact and has a factorisation of each morphism into a strong epimorphism followed by a monomorphism then each congruence in E is the kernel pair of its coequaliser. For example, this occurs if E contains a small strongly generating set of objects so that E is well-powered.

The following result is then an immediate consequence of the $\mathcal{V}$ -Yoneda lemma and the definition of sheaf (using the fact that a coproduct can be expressed as the directed colimit of its finite subcoproducts),

(2.7) Lemma Suppose E is completely exact and V is completely regular. Then the embedding of E into $Sh(E, V)$ preserves:

- (i) coequalisers of congruences;
- (ii) small coproducts with monomorphic coprojections. $\square$

(2.8) Definition E is called V -admissible if each functor from E^{op} to V is a strong quotient of a coproduct of representable functors. $\square$

We are now able to prove the following generalised additive version of Giraud's theorem characterising topoi (cf. [4] §0.4). We assume that V is a cocomplete additive completely regular finitely presentable closed category in which all small coproducts have monomorphic coprojections (for example, a convolution of the form $[M, Ab]$ where M is a small additive symmetric monoidal category).

(2.9) Theorem. Let E be a V -admissible V -category. Then E is V -equivalent to a V -topos if and only if the following three conditions are satisfied:

- (i) E is completely exact;
- (ii) E has a small strongly generating set;
- (iii) E has monomorphic coprojections into coproducts.

Proof Necessity of the conditions follows from the hypotheses on V and the definition of a V -topos. For sufficiency it remains to show that each sheaf in $Sh(E, V)$ is representable since, by Comparison lemma (2.4), we have $Sh(E, V)$ is V -equivalent to $Sh(C, V)$ where C is the closure in E of the given small strongly generating set under

finite limits and subobjects. Assume firstly that F is a subfunctor of a representable sheaf. Since E is assumed to be $\mathcal{V}$ -admissible we have, by (2.4), that F is a strong quotient of a coproduct

$$\sum_{\lambda \in \Lambda} E(-, IC_{\lambda})$$

where $C_{\lambda} \in \mathcal{C}$ for all $\lambda \in \Lambda$. Thus F is a strong quotient in $Sh(E, \mathcal{V})$ of a representable $E(-, E)$ by Lemma (2.7)(ii) and the assumption on coproducts in E . It now follows from Lemma (2.7)(i) that F is itself representable. Next assume F in $Sh(E, \mathcal{V})$ is arbitrary. The kernel pair of the quotient

$$\sum_{\lambda \in \Lambda} E(-, IC_{\lambda}) \longrightarrow F$$

is then a subfunctor of a representable (by Lemma (2.7)(ii)) and thus is representable. Hence, by Lemma (2.7)(i) F is itself representable, as required. $\square$

For examples of $\mathcal{V}$ -admissible categories in this context, first recall that a symmetric monoidal closed category M is called compact if the canonical morphism

$$[m, I] \otimes n \longrightarrow [m, n]$$

is an isomorphism for all objects m and n of M .

(2.10) Proposition. If M is a small additive compact closed category and V is the convolution $[M, Ab]$ then every finitely complete V -category E is V -admissible.

Proof. It suffices that $M(m, -) \otimes E(-, E)$ be representable for all objects m in M and E in E . However, by the V -Yoneda lemma, we have the following isomorphism:

$$\begin{aligned}
 M(m, -) \otimes E(-, E) &\cong E(-, E)[m, -] \\
 &\cong E(-, E)([m, I] \otimes -) \\
 &\cong [M([m, I], -), E(-, E)] \\
 &\cong E(-, [M([m, I], -), E]). \quad \square
 \end{aligned}$$

We remark, in conclusion, that there is a corresponding generalised form of Giraud's original *Set*-based theorem (cf. [4] §0.4) for the case where V is a convolution $[M, Set]$ with M a small symmetric monoidal category. The details of the proof are precisely analogous to those of the additive case, except that we replace small directed colimits by small disjoint coproducts.

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